

Certificate_Validator

Complete Post-Cleanup Project Structure & Architecture Report

Status Overview: All 9 unused/unwanted files have been removed. The application builds cleanly with 0 errors (Vite build output verified). This document outlines the complete directory structure after cleanup.

Summary of Deleted Unused Files

Filename / Path	Reason for Removal
DEMO_SCRIPT.md	Demo preparation document; not used by app
docs/templates/certificate_gallery.html	Standalone preview HTML; not referenced
frontend/src/components/OfflineIndicator.jsx	Unused component; removed import from App.jsx
frontend/src/components/wallet/AICareerInsightsCard.jsx	Card removed from Wallet Dashboard
frontend/src/components/wallet/CareerReadinessCard.jsx	Card removed from Wallet Dashboard
frontend/src/components/wallet/ShareModal.jsx	Deprecated stub file returning null
frontend/src/components/wallet/VirtualizedCertGrid.jsx	Unused virtual grid component
frontend/src/components/wallet/WalletStats.jsx	Component unused (API fn handled directly)
frontend/src/components/wallet/WalletToolbar.jsx	Unused toolbar component

Complete Project Directory Structure (Post-Cleanup)

1. Root Workspace & Environment

```
Certificate_Validator/  
|-- .gitignore  
|-- package.json  
|-- package-lock.json  
`-- README.md
```

2. Documentation Directory (docs/)

```
Certificate_Validator/docs/
|-- screenshots/
|   |-- certificate-template.jpg
|   |-- fraud-analysis-modal.png
|   |-- platform-overview.png
|   |-- verified-status-stamp.png
|-- templates/
|   |-- 01_graduation.jpg
|   |-- 02_distinction.jpg
|   |-- 03_merit.jpg
|   |-- 04_academic_excellence.jpg
|   |-- 05_course_completion.jpg
|   |-- 06_internship.jpg
|   |-- 07_project.jpg
|   |-- 08_bonafide.jpg
|   |-- 09_participation.jpg
```

3. Backend Core & Database (backend/)

```
Certificate_Validator/backend/
|-- .env
|-- .env.example
|-- .gitignore
|-- database.sqlite
|-- database.sqlite-shm
|-- database.sqlite-wal
|-- package.json
|-- package-lock.json
|-- server.js
|-- config/
|   |-- db.js
|-- middleware/ (3 middleware modules)
|   |-- auth.js
|   |-- rateLimiter.js
|   |-- upload.js
|-- models/ (4 data models)
|   |-- chatHistoryModel.js
|   |-- fraudAnalysisModel.js
|   |-- passportModel.js
|   |-- templateModel.js
|-- tests/ (4 backend unit test suites)
|   |-- crypto.test.js
|   |-- disclosure.test.js
|   |-- revocation.test.js
|   |-- verification.test.js
|-- uploads/
|   |-- .gitkeep (Dynamically generated QR assets)
|-- utils/ (3 backend helper utilities)
|   |-- auditLogger.js
|   |-- blockchain.js
|   |-- crypto.js
```

4. Backend Controllers, Routes & Services

```
Certificate_Validator/backend/
|-- controllers/ (15 API controllers)
|   |-- analyticsController.js       |-- auditController.js
|   |-- authController.js           |-- blockchainController.js
|   |-- certificateController.js     |-- chatAssistantController.js
|   |-- disclosureController.js     |-- fraudDetectionController.js
|   |-- goalController.js           |-- passportController.js
|   |-- revocationController.js     |-- templateController.js
|   |-- universityController.js     |-- verificationController.js
|   |-- walletController.js
|-- routes/ (15 API route handlers)
|   |-- analyticsRoutes.js          |-- auditRoutes.js
|   |-- authRoutes.js              |-- blockchainRoutes.js
|   |-- certificateRoutes.js        |-- chatAssistantRoutes.js
|   |-- disclosureRoutes.js         |-- fraudRoutes.js
|   |-- goalRoutes.js              |-- passportRoutes.js
|   |-- revocationRoutes.js         |-- templateRoutes.js
|   |-- universityRoutes.js         |-- verificationRoutes.js
|   |-- walletRoutes.js
|-- services/ (9 core AI & system services)
|   |-- aiRiskScoringService.js     |-- chatAssistantService.js
|   |-- contextBuilder.js           |-- fraudDetectionService.js
|   |-- llmProvider.js              |-- ocrService.js
|   |-- passportService.js          |-- promptBuilder.js
|   |-- templateService.js
```

5. Frontend Core, Config & Public Assets

```
Certificate_Validator/frontend/
|-- .gitignore
|-- eslint.config.js
|-- index.html
|-- package.json
|-- package-lock.json
|-- README.md
|-- vite.config.js
|-- public/
|   |-- favicon.svg
|   |-- icons.svg
|   |-- pwa-icon-192.jpg
|   |-- pwa-icon-512.jpg
|-- src/
|   |-- App.jsx
|   |-- index.css
|   |-- main.jsx
|   |-- styles/
|   |   |-- wallet.css
|   |-- api/ (10 API integration modules)
|   |   |-- analytics.js      |-- audit.js
|   |   |-- blockchain.js    |-- chat.js
|   |   |-- client.js        |-- disclosure.js
|   |   |-- fraud.js         |-- passport.js
|   |   |-- templateApi.js   |-- wallet.js
|   |-- hooks/ (3 custom hooks)
|   |   |-- useAuthProvider.js |-- useHeaderHeight.js
|   |   |-- useReveal.js
|   |-- utils/ (8 utility helpers)
|   |   |-- certificateCategory.js |-- certificatePdf.js
|   |   |-- excelParser.js         |-- keyCache.js
|   |   |-- offlineCrypto.js       |-- qrDecoder.js
|   |   |-- revocationCache.js     |-- walletStore.js
```

6. Frontend Application Pages (frontend/src/pages/)

```
Certificate_Validator/frontend/src/pages/ (19 Pages)
|-- AuditLog.jsx      |-- BlockchainExplorer.jsx
|-- DigitalSkillPassport.jsx |-- Home.jsx
|-- PublicDisclosureView.jsx |-- PublicSkillPassport.jsx
|-- Register.jsx      |-- StudentAnalytics.jsx
|-- StudentDashboard.jsx |-- StudentLogin.jsx
|-- StudentRegister.jsx |-- TemplateManager.jsx
|-- UniversityAnalytics.jsx |-- UniversityDashboard.jsx
|-- UniversityLogin.jsx   |-- UniversityRegister.jsx
|-- VerificationAnalytics.jsx |-- Verifier.jsx
|-- WalletDashboard.jsx
```

7. Frontend Components (frontend/src/components/)

```
Certificate_Validator/frontend/src/components/
|-- AuditLogDecorations.jsx      |-- AuthBackgroundDecorations.jsx
|-- CertificateTemplate.jsx      |-- FraudAnalysisModal.jsx
|-- GuidedTourModal.jsx         |-- StudentDashboardDecorations.jsx
|-- UniversityDashboardDecorations.jsx
|-- VerifierBackgroundDecorations.jsx
|-- AIChat/ (5 AI chat components)
|   |-- ChatBubble.jsx          |-- ChatWindow.jsx
|   |-- FloatingAIButton.jsx    |-- SuggestedQuestions.jsx
|   |-- TypingIndicator.jsx
|-- decorations/ (5 dashboard decoration graphics)
|   |-- BlockchainExplorerDecorations.jsx
|   |-- StudentAnalyticsDecorations.jsx
|   |-- UniversityAnalyticsDecorations.jsx
|   |-- VerificationAnalyticsDecorations.jsx
|   |-- WalletDashboardDecorations.jsx
|-- motion/ (4 animation utilities)
|   |-- CountUp.jsx             |-- index.js
|   |-- RevealOnScroll.jsx      |-- SkeletonCard.jsx
|-- templates/ (11 certificate templates)
|   |-- AcademicExcellenceTemplate.jsx |-- BonafideTemplate.jsx
|   |-- CategoryCertificateTemplate.jsx |-- CourseCompletionTemplate.jsx
|   |-- DistinctionTemplate.jsx      |-- GraduationTemplate.jsx
|   |-- InternshipTemplate.jsx       |-- MeritTemplate.jsx
|   |-- ParticipationTemplate.jsx    |-- ProjectTemplate.jsx
|   |-- TemplateSelector.jsx
|-- wallet/ (7 student wallet components)
|   |-- AchievementTimeline.jsx      |-- CertDetailModal.jsx
|   |-- LearningGoalTracker.jsx     |-- PortfolioLinksCard.jsx
|   |-- WalletCertCard.jsx          |-- WalletDetailPane.jsx
|   |-- WalletIndexList.jsx
```